



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**FINAL EXAMINATION
SEMESTER III, SESSION 2020/2021
PART I**

COURSE CODE	:	FSPC 0034
COURSE NAME	:	CHEMISTRY II
PROGRAMME	:	ASASI UTMSPACE
DURATION	:	1 HOUR 30 MINUTES
DATE	:	15 APRIL 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. THERE ARE **FOUR (4)** QUESTIONS. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (8) printed pages including this page.

QUESTION 1 (10 MARKS)

- a) Calcium is reacted with chlorine to form calcium chloride. The enthalpy changes associated with the reaction are given below.

Enthalpy change of atomization of calcium = $+121 \text{ kJ mol}^{-1}$

First ionization energy of calcium = $+590 \text{ kJ mol}^{-1}$

Second ionization energy of calcium = $+1150 \text{ kJ mol}^{-1}$

Enthalpy change of atomization of chlorine = $+121 \text{ kJ mol}^{-1}$

Electron affinity of chlorine = -364 kJ mol^{-1}

Enthalpy change of formation of calcium chloride = -795 kJ mol^{-1}

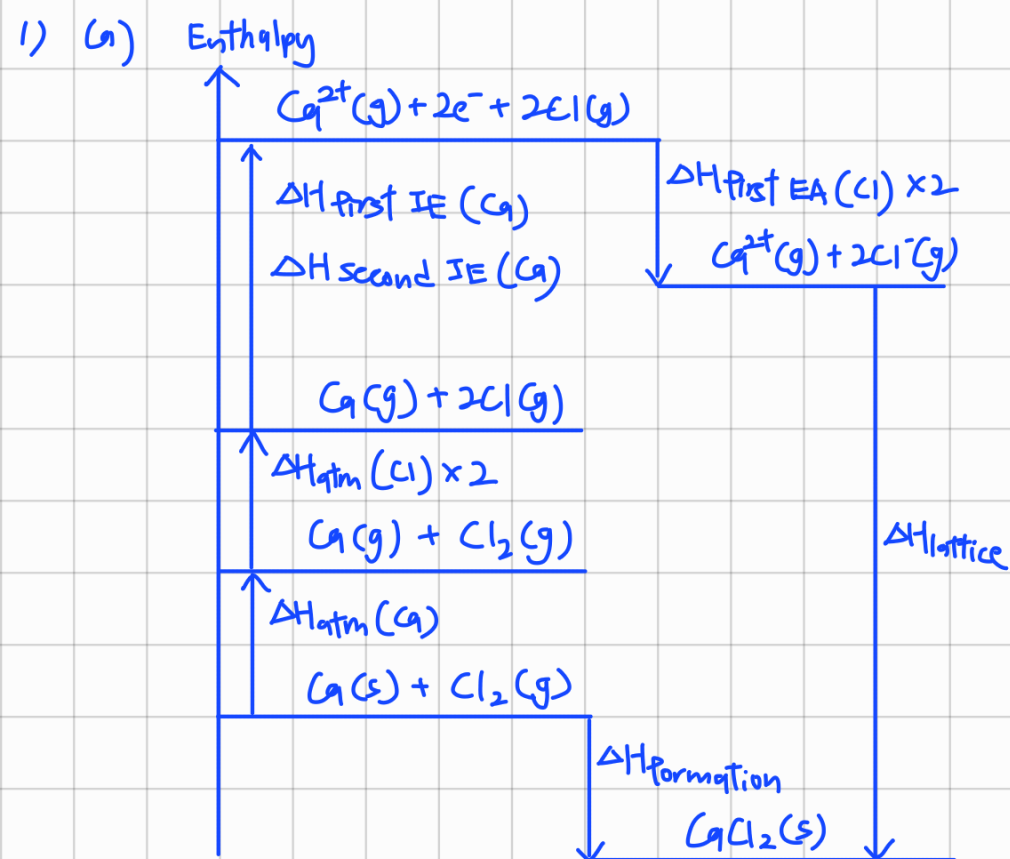
- i. Based on the information given above, construct and draw Born-Haber cycle. Please include all the energies/enthalpy changes and chemical equations involved.
- ii. Determine the Born-Haber cycle equation for calcium chloride and calculate the lattice energy of $\text{CaCl}_2 (\text{s})$.

(5 + 2 marks)

- b) The standard enthalpies of combustion of 1-propanol, carbon and hydrogen are $-2010 \text{ kJ mol}^{-1}$, -394 kJ mol^{-1} and -286 kJ mol^{-1} respectively.

- i. Write the thermochemical equation for the combustion of liquid 1-propanol.
- ii. Determine the standard enthalpy equation of this reaction. Then, calculate the standard enthalpy of formation of 1-propanol.

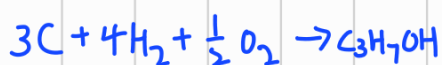
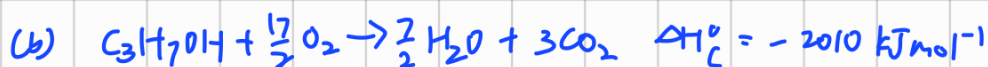
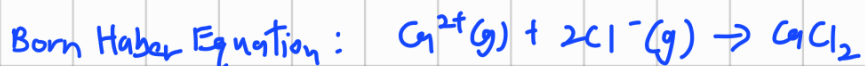
(1 + 2 marks)



$$\Delta H_{\text{formation}} = H_{\text{lattice}} + H_{\text{atm}}(\text{Ca}) + H_{\text{atm}}(\text{Cl}) + H_{\text{IE}}(\text{Ca}) + H_{\text{EA}}(\text{Cl})$$

$$-795 = \Delta H_{\text{lattice}} + 121 + 121 + 590 + 1150 + (-364)$$

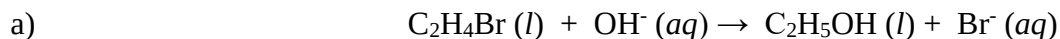
$$\Delta H_{\text{lattice}} = -2413 \text{ kJ mol}^{-1}$$



$$\Delta H_f = [3(-394) + 4(-286)] - (-2010)$$

$$= -2326 + 2010$$

$$= -316 \text{ kJ/mol}$$

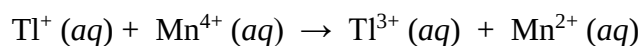
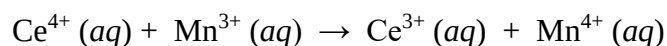
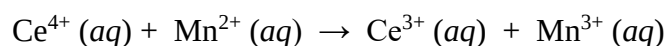
QUESTION 2 (10 MARKS)

The equation above describes the reaction between ethyl bromide ($\text{C}_2\text{H}_4\text{Br}$) and hydroxide ion in ethyl alcohol at 330 K, in which ethyl bromide is second order while hydroxide ion is zero order. When $[\text{C}_2\text{H}_4\text{Br}]$ is 0.1022 M and $[\text{OH}^-]$ is 0.233 M, the rate of disappearance of ethyl bromide is $1.7 \times 10^{-7} \text{ M/s}$.

- i. Based on the information given above, calculate the rate constant value of the reaction.
- ii. Determine the unit for the rate constant.

(2 + 1 marks)

- b) An example of a multi-steps catalytic reaction is shown below.



- i. Based on the chemical equations, state the catalyst used in the reaction.
- ii. Determine the overall equation for the reaction.
- iii. Given that the rate for the catalyzed reaction is first order with respect to Ce^{4+} and first order with respect to Mn^{2+} . Which of the steps is the rate limiting step?

(1 + 1 + 1 marks)

- c) Consider a reaction between $\text{CH}_3\text{COOC}_2\text{H}_5$ and OH^- , it was found that the order of reaction is first order with respect to $\text{CH}_3\text{COOC}_2\text{H}_5$ and second order with respect to OH^- . Given that the activation energy for the reaction is $2.1 \times 10^2 \text{ kJ/mol}$ at 25°C , the rate of reaction is $6.5 \times 10^{-4} \text{ Ms}^{-1}$ when $[\text{CH}_3\text{COOC}_2\text{H}_5] = 0.7 \text{ M}$ and $[\text{OH}^-] = 0.5 \text{ M}$. Determine the rate of reaction at a temperature of 40°C .

(4 marks)

2) (a) (i) $k = [\text{C}_2\text{H}_4\text{Br}]^2 [\text{OH}^-]^0$, $[\text{C}_2\text{H}_4\text{Br}] = 0.1022 \text{ M}$, $[\text{OH}^-] = 0.233 \text{ M}$

$$k = (0.1022)^2 (0.233)^0$$

$$= 1.04 \times 10^{-2}$$

(ii) s^{-2}

(b) (i) catalyst: Mn^{2+}



(iii) step 2

(c) $6.5 \times 10^{-4} = k [\text{CH}_3\text{COOC}_2\text{H}_5]^1 [\text{OH}^-]^2$ $E_A = 2.1 \times 10^2 \text{ kJ/mol}$ $T_1 = 298 \text{ K}$ $T_2 = 313 \text{ K}$

$$6.5 \times 10^{-4} = k(0.7)(0.5)^2$$

$$k = \frac{6.5 \times 10^{-4}}{0.7 \times 0.5^2}$$

$$= 3.714 \times 10^{-3}$$

$$\ln\left(\frac{k_2}{3.714 \times 10^{-3}}\right) = \frac{2.1 \times 10^2}{8.314} \left(\frac{1}{313} - \frac{1}{298}\right)$$

$$R = k(0.7)(0.5^2)$$

$$= 3.698 \times 10^{-3} (0.175)$$

$$= 6.472 \times 10^{-4} \text{ M s}^{-1}$$

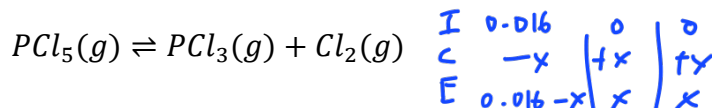
$$\frac{k_2}{3.714 \times 10^{-3}} = e^{-4.06 \times 10^{-3}}$$

$$k_2 = 3.714 \times 10^{-3} (9.959 \times 10^{-1}) = 3.698 \times 10^{-3}$$

QUESTION 3 (10 MARKS)

m.m of PCl₅ : 208.24, mass PCl₅ : 6.67g
mol = $\frac{6.67}{208.24} = 3.2 \times 10^{-2} \text{ mol}$, $[\text{PCl}_5] = \frac{3.2 \times 10^{-2}}{2} = 1.6 \times 10^{-2} \text{ M}$

- a) Phosphorus pentachloride has been used as a chlorinating reagent in the manufacturing of antibiotics as well as for the manufacturing of electrolytes for lithium-ion batteries. At 250°C, phosphorus (V) chlorate dissociates according to the equation:



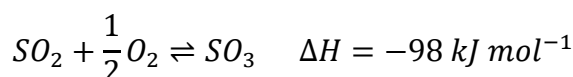
When 6.67 g of gaseous PCl₅ was placed in a 2.0 L vessel at 250°C, 60% of PCl₅ underwent dissociation to form PCl₃ and Cl₂. Determine the equilibrium constant, K_c for this reaction.

0.016 × 60% = 0.0096
x = 0.0096 M

$$K_c = \frac{0.0096^2}{0.016 - 0.0096} \quad (4 \text{ marks})$$

= 0.0144 M

- b) The critical step in the manufacture of sulphuric acid is the oxidation of sulphur dioxide.



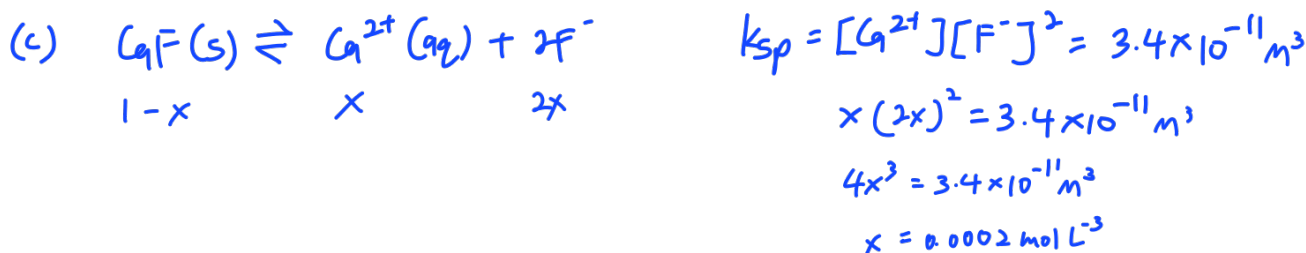
Explain why the reaction is carried out at 425°C to make the process as economic as possible.

(3 marks)

- c) Calcium fluoride (CaF₂) which often recognized as fluorite or fluorspar is the major source of fluorine in manufacture of hydrogen fluoride and calcium sulfate. Determine the solubility (in g L⁻¹) of calcium fluoride (CaF₂) in water.

(K_{sp} of CaF₂ = 3.4 × 10⁻¹¹ mol³ L⁻⁹)

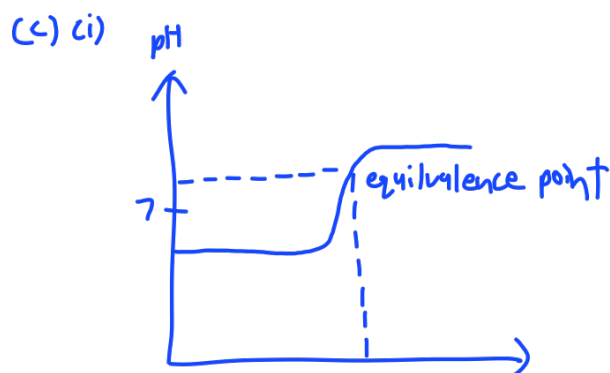
(3 marks)

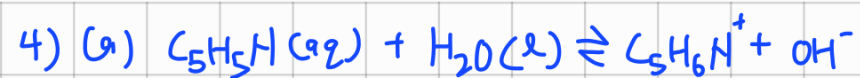


solubility = 0.0002 (40 + 19)
= 0.012 g L⁻³

QUESTION 4 (10 MARKS)

- a) Pyridine, C_5H_5N is a base and will react with water.
- Write the chemical equation for the reaction of pyridine with water.
 - Using pyridine as the base, identify the other components to make a buffer solution.
- (1 + 1 marks)
- b) i. If hydrogen fluoride, HF has a pH of 5.6, determine the concentration of H^+ .
- ii. If HF is reacted with methylamine, predict the pH at equivalence point.
 K_a of HF = 6.8×10^{-4}
 K_b of methylamine = 4.4×10^{-4}
- (1 + 1 marks)
- c) Benzoic acid, $HC_7H_5O_2$ is commonly used as additive for food preservation, and various commercial product. A 0.250 M solution of benzoic acid (titrant) is titrated to 20 mL 0.125 M barium hydroxide, $Ba(OH)_2$ solution.
 (Given $K_a = 6.3 \times 10^{-5}$, $K_w = 1 \times 10^{-14}$)
- Plot the expected titration curve of the acid-base titration.
 - Calculate the pH of the solution after the addition of 20.00 mL 0.250 M benzoic acid solution.
- (2 + 4 marks)





salt of conjugate acid NaC_5H_6N

(b) $pH = 5.6 = -\log[H^+]$

$[H^+] = 10^{-5.6}$

$= 2.512 \times 10^{-6} M$

Since $K_a \approx K_b$, pH is almost equal to 7

(c) ① $\frac{n_a V_a}{M_b V_b} = \frac{a}{b}$

$\frac{0.25(V_a)}{0.125(20)} = \frac{2}{1}$

V_{acid} at equivalence point = 20 mL

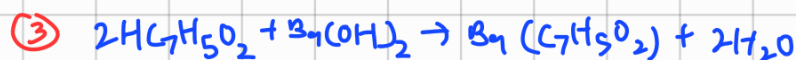
② mol of $Ba(OH)_2 = 2.5 \times 10^{-3}$

mol of $HC_7H_5O_2 = 5 \times 10^{-3}$

⑤ $K_b = \frac{K_w}{K_a}$
 $= \frac{1.0 \times 10^{-14}}{6.3 \times 10^{-5}}$
 $= 1.587 \times 10^{-10}$

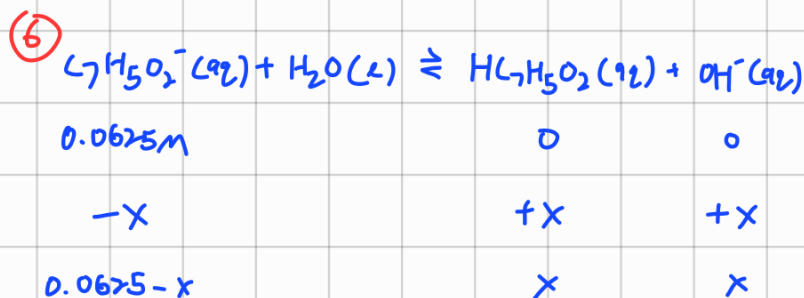
⑦ $1.587 \times 10^{-10} = \frac{x^2}{0.0625}$
 $x = 3.15 \times 10^{-6} M$

⑧ $pOH = -\log(3.15 \times 10^{-6})$ $pH = 14 - 5.5$
 $= 5.5$ $= 8.5$



B	5×10^{-3}	2.5×10^{-3}	0
A	$-2(2.5 \times 10^{-3})$	-2.5×10^{-3}	$+2.5 \times 10^{-3}$
A	0	0	2.5×10^{-3}

④ $[C_7H_5O_2^-] = \frac{2.5 \times 10^{-3}}{0.02 + 0.02}$
 $= 0.0625 M$



Appendix 1 : THE PERIODIC TABLE

1	2	13	14	15	16	17	18
IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA
1 H 1.01	2 He 4.00	5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
3 Li 6.94	4 Be 9.01	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
11 Na 22.99	12 Mg 24.31	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
19 K 39.1	20 Ca 40.08	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.6	53 I 126.9	54 Xe 131.29
37 Rb 85.47	38 Sr 87.62	81 Tl 204.4	82 Pb 207.2	83 Bi 209	84 Po (209)	85 At (210)	86 Rn (222)
55 Cs 132.9	56 Ba 137.3	89 Ac[^] (227)	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)
87 Fr (223)	88 Ra (226)	101 La[*] 138.9	102 Ce 140.1	103 Pr 140.9	104 Nd 144.2	105 Pm (145)	106 Sm 150.4
		107 Re 186.2	108 Os 190.2	109 Ir 192.2	110 Pt 195.1	111 Au 197.0	112 Hg 200.6
		109 Mt (268)	110 Ds (271)	111 Rg (272)	112 Uut (285)	113 Uuq (289)	114 Uub (293)
		113 Uut (285)	114 Uuq (289)	115 Uub (293)	116 Uut (297)	117 Uuq (301)	118 Uub (305)
		115 Uub (293)	116 Uut (297)	117 Uuq (301)	118 Uub (305)	119 Uut (309)	120 Uuq (313)
		117 Uuq (301)	118 Uub (305)	119 Uut (309)	120 Uuq (313)	121 Uub (317)	122 Uut (321)
		119 Uub (309)	120 Uuq (313)	121 Uub (317)	122 Uut (321)	123 Uuq (325)	124 Uub (329)
		121 Uub (309)	122 Uut (321)	123 Uuq (325)	124 Uub (329)	125 Uut (333)	126 Uuq (337)
		123 Uuq (325)	124 Uub (329)	125 Uut (333)	126 Uuq (337)	127 Uub (341)	128 Uut (345)
		125 Uub (309)	126 Uut (321)	127 Uuq (325)	128 Uub (329)	129 Uut (333)	130 Uuq (337)
		127 Uuq (325)	128 Uub (329)	129 Uut (333)	130 Uuq (337)	131 Uub (341)	132 Uut (345)
		129 Uub (309)	130 Uut (321)	131 Uuq (325)	132 Uub (329)	133 Uut (333)	134 Uuq (337)
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		133 Uuq (325)	134 Uub (329)	135 Uut (333)	136 Uuq (337)	137 Uub (341)	138 Uut (345)
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		195 Uub (309)	196 Uut (321)	197 Uuq (325)	198 Uub (329)	199 Uut (333)	200 Uuq (337)
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		201 Uub (309)	202 Uut (321)	203 Uuq (325)	204 Uub (329)	205 Uut (333)	206 Uuq (337)
		203 Uuq (325)	204 Uub (329)	205 Uut (333)	206 Uuq (337)	207 Uub (341)	208 Uut (345)
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		209 Uub (309)	210 Uut (321)	211 Uuq (325)	212 Uub (329)	213 Uut (333)	214 Uuq (337)
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		249 Uub (309)	250 Uut (321)	251 Uuq (325)	252 Uub (329)	253 Uut (333)	

Appendix 2 : LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Appendix 3 : STANDARD REDUCTION POTENTIAL (E°_{Red})

$\text{PbO}_2(\text{s}) + \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O}(\ell)$	+1.69
$\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow$	$\text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.51
$\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow$	$\text{Au}(\text{s})$	+1.50
$\text{ClO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 8\text{e}^- \rightarrow$	$\text{Cl}^-(\text{aq}) + 4\text{H}_2\text{O}(\ell)$	+1.39
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightarrow$	$2\text{Cl}^-(\text{aq})$	+1.36
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14\text{H}^+(\text{aq}) + 6\text{e}^- \rightarrow$	$2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\ell)$	+1.33
$2\text{HNO}_3(\text{aq}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow$	$\text{N}_2\text{O}(\text{g}) + 3\text{H}_2\text{O}(\ell)$	+1.30
$\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow$	$2\text{H}_2\text{O}(\ell)$	+1.23
$\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O}(\ell)$	+1.22
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightarrow$	$2\text{Br}^-(\text{aq})$	+1.07
$\text{Hg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Hg}(\ell)$	+0.85
$\text{ClO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{Cl}^-(\text{aq}) + 2\text{OH}^-(\text{aq})$	+0.84
$\text{Ag}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s})$	+0.80
$\text{NO}_3^-(\text{aq}) + 2\text{H}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\ell)$	+0.80
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \rightarrow$	$\text{Fe}^{2+}(\text{aq})$	+0.77
$\text{O}_2(\text{g}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{O}_2(\ell)$	+0.70
$\text{I}_2(\text{s}) + 2\text{e}^- \rightarrow$	$2\text{I}^-(\text{aq})$	+0.54
$\text{O}_2(\text{g}) + 2\text{H}_2\text{O}(\ell) + 4\text{e}^- \rightarrow$	$4\text{OH}^-(\text{aq})$	+0.40
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cu}(\text{s})$	+0.34
$\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{SO}_3(\text{aq}) + \text{H}_2\text{O}(\ell)$	+0.17
$\text{Sn}^{4+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Sn}^{2+}(\text{aq})$	+0.15
$\text{S}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{S}(\text{aq})$	+0.14
$\text{AgBr}(\text{s}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s}) + \text{Br}^-(\text{aq})$	+0.07
$2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_{2(\text{g})}$	0.00
$\text{Pb}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Sn}(\text{s})$	-0.14
$\text{AgI}(\text{s}) + \text{e}^- \rightarrow$	$\text{Ag}(\text{s}) + \text{I}^-(\text{aq})$	-0.15
$\text{Ni}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ni}(\text{s})$	-0.26
$\text{Co}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Co}(\text{s})$	-0.28
$\text{PbSO}_4(\text{s}) + 2\text{e}^- \rightarrow$	$\text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq})$	-0.36
$\text{Se}(\text{s}) + 2\text{H}^+(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{H}_2\text{Se}(\text{aq})$	-0.40
$\text{Cd}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+}(\text{aq}) + \text{e}^- \rightarrow$	$\text{Cr}^{2+}(\text{aq})$	-0.41
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Fe}(\text{s})$	-0.45
$\text{NO}_2^-(\text{aq}) + \text{H}_2\text{O}(\ell) + \text{e}^- \rightarrow$	$\text{NO}(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.46
$\text{Ag}_2\text{S}(\text{s}) + 2\text{e}^- \rightarrow$	$2\text{Ag}(\text{s}) + \text{S}^{2-}(\text{aq})$	-0.69
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{H}_2(\text{g}) + 2\text{OH}^-(\text{aq})$	-0.83
$\text{Cr}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Cr}(\text{s})$	-0.91
$\text{Se}(\text{s}) + 2\text{e}^- \rightarrow$	$\text{Se}^{2-}(\text{aq})$	-0.92
$\text{SO}_4^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) + 2\text{e}^- \rightarrow$	$\text{SO}_3^{2-}(\text{aq}) + 2\text{OH}^-(\text{aq})$	-0.93
$\text{Al}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow$	$\text{Al}(\text{s})$	-1.66
$\text{Mg}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Mg}(\text{s})$	-2.37
$\text{Na}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ca}(\text{s})$	-2.87
$\text{Ba}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow$	$\text{Ba}(\text{s})$	-2.91
$\text{Li}^+(\text{aq}) + \text{e}^- \rightarrow$	$\text{Li}(\text{s})$	-3.04